
Forecasting Is Not World Modeling: Accuracy, Action Fidelity, and an Evaluation Matrix for a Task-Critical Thermal Process

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Abstract

1 Task-critical industrial processes demand more from a temporal model than accurate
2 forecasts. A model promoted to the inner loop of a controller must respond to
3 candidate actions with the right sign, magnitude, and transient shape, and it must
4 know when it cannot. The question arose on a live unit where a monitoring forecast
5 had been wired into the control chain: a 180-step-ahead predictor whose 90-second-
6 ahead prediction corrects the temperature setpoint as differential feedforward. That
7 system forecasts well, and it barely responds to its own valves. We study this gap
8 on a once-through boiler in dry operation, using 82 days of dual-side field data. A
9 deep forecaster attains 0.7–0.9 °C MAE at 18 steps, yet its valve-step responses
10 span 0.00004–0.040 °C with unstable direction across folds, valves, and seeds—
11 functionally disabling the action channel. A physics-structured world model trades
12 accuracy for an action channel that a first training exposed as directionally wrong
13 at 60 steps—an inversion we localize to one feedback term—and that a contracted
14 repair batch then corrected: on the repaired seed the direction holds on 100% of
15 windows (mean -0.228 °C per $+5\%$ valve step), with the remaining seeds pending.
16 Both findings are produced by a discrimination matrix: an identifiability gate on
17 same-type synthetic data, per-layer structural ablations, and an action-blindness
18 and shuffle-controlled leakage audit—a protocol whose own false trigger we report
19 and repair in the loop. Finally we locate the physics model’s accuracy cost: it
20 approaches persistence at the terminal stage (0.16 vs. 0.09 °C one-step MAE,
21 beating persistence at 18 steps) while the upstream stage fails $38\times$ worse than
22 persistence, and the earlier monotone regime bias collapses to ± 0.2 °C—a training
23 artefact, not a parameter-count one. Our results support verifiability, not accuracy,
24 as the qualifying property of a world model, and a two-tier architecture in which a
25 learned proposer is bounded by a certifiable reduced-order model. Matrix verdicts
26 are single-side (side A); the side-B matrix is deferred to a later stage.

27 1 Introduction

28 A temporal model that forecasts well is not automatically a world model. World models must maintain
29 a state, imagine futures under interventions, and support decisions [5]. Closed-loop identification
30 warned decades ago that passively learned dynamics absorb the confounded structure of the controller
31 that generated the data [3], and recent evidence shows time-series foundation models collapsing
32 under feedback despite strong open-loop accuracy [9]. This paper is a case study of that failure being
33 encountered in the field, chased to its root, and turned into a reusable protocol.

34 **Where we started.** On a once-through boiler at an industrial plant, a monitoring forecast was
35 extended toward closed-loop use. The model predicts 180 steps ahead, but the control system

36 accepts a single connection point; the 90-second-ahead prediction—the horizon with the best one-
37 step accuracy and the most reliable action response—is wired into the setpoint-correction loop as
38 differential feedforward, realizing lead control ahead of the plant’s multi-minute response lag. The
39 system forecasts well. It also barely reacts to its own valves: changing the spray-valve positions
40 leaves the prediction almost unchanged, and where a response does appear its direction is unreliable.
41 For a feedforward loop whose physical lever *is* the valve, this is disqualifying.

42 **Why not learn a world model the usual way.** In robot control and autonomous driving, the standard
43 remedies are interaction-based: online reinforcement learning explores the action space, offline RL
44 reuses logged interaction or a simulator [5]. Neither is available here. Online trial is barred by safety
45 class: a superheater excursion trips the unit, and exploratory dithering of spray valves—the discrete
46 probing RL would want—is operationally unacceptable. Offline training in simulation is barred
47 by a transfer gap the other way: the boiler is strongly coupled, disturbance-rich, and time-varying,
48 and a simulator faithful enough that controllers learned inside it transfer to the plant does not exist.
49 Closed-loop operating records are the only admissible data. The problem this paper addresses is
50 therefore sharper than generic world-model evaluation: *how to identify the causal sign and magnitude*
51 *of an action channel from data generated under feedback, when the controller systematically couples*
52 *the valve to the very temperature it drives.* This common-cause structure is precisely what the
53 diagnosis below unpacks.

54 **The diagnosis.** Our earlier diagnosis quantified the anomaly in three layers. First, representation: the
55 forecast did not ignore the valves by architecture—a model with a zeroed action pathway still reached
56 0.808 °C rollout MAE, and switching from differential to absolute valve inputs enlarged the learned
57 response by a factor of 32–130. One-step metrics are blind to it by construction: the plant’s true
58 spray-to-outlet lag exceeds 120 s, so a correct one-step response is nearly zero. Second, timescale:
59 single-step sign regularization therefore not only failed, it compressed a 120 s physical lag into a
60 spurious 10 s response. Third, confounding: the controller opens the spray valve when temperature
61 rises, so “valve-open $\leftrightarrow$ temperature-rising” is a statistical fact of the data and a physical falsehood.

62 **The illusion.** A constrained gray-box branch appeared to fix the direction: its action response was
63 sign-consistent across the evaluation. A 1 s-resolution audit exposed what it had learned. Setpoint
64 changes are rare (0.76% of steps) and the exogenous valve signal is rarer still (0.2% of steps; 99.8%
65 of valve motion is feedback-driven). The branch’s “correct” sign was closed-loop *tracking*: the
66 setpoint rises, the controller closes the spray valve, the temperature rises—the model learned to track
67 the setpoint through the controller, not the plant’s physical valve-to-temperature pathway. Direction
68 had been faked once. Every claim made for the physics line in this paper is therefore gated by an
69 audit, not by its architecture.

70 **Two lines, each failing alone.** The pure forecasting line (a high-capacity direct forecaster with an
71 action channel) predicts held-out trajectories well but its valve responses span 0.00004–0.040 °C
72 with direction drifting across folds, valves, and seeds. The pure physics line (a lumped-enthalpy
73 ODE, descended from the constrained branch, with a residual closure) responds on the right scale but
74 instantaneously—the plant’s response builds over minutes—and its accuracy is large and initial-state
75 dependent, because the field data carry no reliable spray-flow, wall-temperature, or enthalpy ground
76 truth. Neither line alone qualifies as an internal model for control.

77 **What follows.** We measure the resulting tension three ways and report it through a discrimination
78 matrix: an identifiability gate, structural ablations, and an action-blindness and leakage audit, run
79 by a frozen protocol with verdicts computed by paired bootstrap and a 2/3 seed rule. The matrix
80 is not a checklist; it is a process. Its verdict-repair loop caught its own leakage probe misfiring
81 and protocolized the correction, and one of its convergence diagnostics demoted a parameter-map
82 research line whose evidence turned out to be a training artefact. Our contributions are (i) an empirical
83 demonstration, on dual-side field data, that forecast accuracy does not transfer to world-model
84 qualification; (ii) the discrimination matrix and the protocolized verdict-repair loop it produced; and
85 (iii) a located diagnosis of the physics model’s accuracy cost, with quantified floors and a convergent
86 repair agenda.

87 2 Testbed and Two Model Lines

88 2.1 The process

89 The unit drives steam through a chain of superheater stages interleaved with two spray attenuators:
90 low-temperature superheater, first spray, platen superheater, second spray, final superheater. Five
91 temperatures along this chain are observed at 10 s resolution, together with boundary channels (steam
92 flow, coal command, separator pressure and temperature, feedwater temperature, outlet pressure,
93 spray flow) and two valve positions. The valves are cross-paired: valve A governs side-B temperatures
94 and vice versa. Data span 82 days per side, all in dry operation. Fast temperature increments are
95 small (0.09–0.45 °C per step), so the process is strongly autoregressive; load-driven swings are orders
96 of magnitude larger.

97 2.2 Line 1: a deep forecaster

98 Line 1 is a direct multi-step forecaster (sequence model over a history window with dedicated heads
99 per horizon step, trained with a Gaussian likelihood), the audited descendant of the deployed predictor
100 with an explicit action channel. It is representative of the pure forecasting line rather than a state
101 of the art: the horizon heads are teacher-forced, not recursive. We state this plainly because the
102 comparison that follows is about what the forecasting objective rewards, and the literature suggests
103 recursive foundation forecasters inherit the same closed-loop failure [9]. Its 18-step MAE on held-out
104 windows is 0.712 (side A) and 0.927 (side B).

105 2.3 Line 2: a physics-structured world model

106 Line 2 embeds the process structure: three lumped-enthalpy segments whose dynamics obey energy
107 conservation with IAPWS-97 property tables, two spray injections with transport delays, metal-
108 temperature coupling, and learnable parameters (segment conductances, evaporation gains, spray
109 efficiencies), in the universal-differential-equation tradition of embedding structure while learning
110 parameters [2, 7]. It descends directly from the constrained gray-box branch of Section 1: that
111 branch’s apparent direction fidelity was exposed as controller tracking, and the deeper enthalpy
112 structure is the response to that exposure. An action-blind closure absorbs unmeasured disturbances
113 from state and whitelisted boundary channels only. A probabilistic observer estimates the initial
114 state. The model rolls out recursively, so any counterfactual branch is a consistent trajectory of
115 the same dynamics. Its 18-step MAE is about 3.9 °C under the amended training budget (side A,
116 seed-averaged; see Section 4).

117 3 The Tension, Measured Three Ways

118 3.1 Action response: one line is functionally blind

119 We probe both models with $\pm 2\%$ valve steps, holding all else fixed, and compare against a data-
120 anchored mixing reference (spray-flow regressions on the field data, $dW/dv = 27.8$ and 70.0 t/h per
121 full travel for the two valves, combined with an energy balance at the spray station; the reference spans
122 0.21 – 0.59 °C for the primary valve and 0.53 – 1.48 °C for the secondary valve per $+2\%$, depending
123 on the steam-flow band). Line 1 responds with mean magnitudes of 0.00004 – 0.040 °C across folds,
124 valves, and directions, with the correct sign in only one of eight fold–valve–direction cells at 3/3
125 consistency—two to four orders of magnitude below the reference, and functionally blind (Fig. 3a).

126 Line 2’s first training exposed a different failure. Its one-step response lands on the reference scale,
127 but partly by construction: the action enters an instantaneous mixing equation re-evaluated at every
128 step, so the model responds at step one whether or not the learned dynamics agree, while the real
129 plant’s one-step response is effectively zero and builds over minutes. At 60 steps the direction broke:
130 only 19–34% of windows ended negative (the direction rule as frozen required 100%), with a mean
131 terminal delta of $+0.20$ to $+0.26$ °C. An ablation localized the inversion to one named term: the
132 wall-rewetting feedback. More spray raises the droplet mass, lowering the dry-out factor, raising the
133 rewetting heat flux into the steam; the learned rewetting gain over-compensated the direct cooling.
134 Zeroing the rewetting coefficient restored the correct sign on 100% of windows (mean -0.44 °C)—a
135 pre-repair diagnosis, not a repair.

The contracted repair batch then changed the model, not the probe: a spray-to-mixing transport-lag state, a sign-and-magnitude contract on the rewetting feedback, and spray-sensitivity priors anchored to the data-fit dW/dv . The retrained seed-0 model holds the correct direction on 100% of windows with a mean terminal response of -0.228 °C per $+5\%$ step, and its 240-step (quasi-steady) response is -0.194 °C per $+2\%$. This remains $2.7\text{--}7.6\times$ below the mixing reference. We read the gap as a distance to a nominal linear upper bound, not as demonstrated distortion of the plant: the reference assumes zero-delay mixing, linear valve-to-flow mapping, and fixed $\Delta h/c_p$, while the real valve is equal-percentage and the data contain only narrow-band micro-adjustments (command changes $>3\%$ are absent), so the identifiable quantity is a workpoint-local gain—consistent with the learned spray efficiency landing at $0.44\times$ its prior. Magnitude fidelity remains unverified pending a position-binned local-gain probe. Seeds 1–2: pending ($<R1-3SEED>$).

3.2 Counterfactual coherence

Beyond steady responses, an internal model must carry physically coherent counterfactual branches. Line 1’s teacher-forced heads cannot propagate a perturbation through time; Line 2’s recursive state does so by construction, and its boundary dependence is whitelisted so that no action leaks into the disturbance path. The audit of that channel is itself part of the story, told in Section 4: the leakage probe that first condemned the model was, on a shuffle control, condemning itself.

3.3 Accuracy: the cost is real, located, and mostly structural

Line 2 pays a $3\text{--}4\times$ aggregate accuracy price (H18 MAE 3.9 °C vs. $0.7\text{--}0.9$ for Line 1), but the price is not uniform across the steam path. At the terminal stage the retrained model is nearly as good as persistence one step ahead (0.16 vs. 0.09 °C MAE) and beats it at 18 steps (1.05 vs. 1.18); at the upstream stage it is $38\times$ worse than persistence (9.4 vs. 0.25 °C), and the model’s learned noise scale confirms it has written this channel off ($\sigma \approx 8.8$ °C). Binning residuals by load shows that the strong monotone regime bias present under the initial training budget ($+2.1$ to $+5.4$ °C across load bins at the terminal stage) collapses to ± 0.2 °C after retraining with the amended budget: level calibration was a training-sufficiency artefact, not a parameter-count artefact. What remains is structural: the upstream channel’s residual swings ± 8.8 °C across load bins with no monotone signature, pointing at the enthalpy–temperature inversion at the first segment (pressure partition assumption) and the anchor coverage (two spray-outlet temperatures float at initialization), and the first spray-outlet channel retains 44% regime-correlated variance, the signature of unresolved spray transients (Fig. 3b,c). The autoregressive diagnosis generalizes: the process creeps ($0.09\text{--}0.45$ °C per step), so large one-step errors are model jumps, not noise.

4 A Discrimination Matrix for World-Model Qualification

The tension above is only meaningful if the measurements are reproducible. We froze an evaluation protocol, run end-to-end by a single runner against a versioned data contract, with verdicts computed by paired bootstrap confidence intervals and a 2/3 seed rule. Table 1 summarizes the side-A verdicts.

D-SYN identifiability gate. The full stack (observer, transition, closure) trains on synthetic data generated by the same architecture and must recover the teacher: 3/3 seeds pass with NLL reductions of 99–136, well above the 30% threshold. Failures on real data therefore reflect structure or data, not an unidentifiable task.

O1 initial-state posterior. A learned posterior must beat the steady-state anchor by at least 5% with confidence. It is MIXED (two of three seeds degrade significantly, one improves), and the precision-weighted hybrid is REJECTED (no seed reaches a significant improvement). Under the amended training budget the steady anchor itself improved enough to erase the posterior’s advantage; the learned posterior’s seed instability remains.

T1 transition structure. Nested arms add closure layers and a latent block against the previous layer. The conservative closure is SUPPORTED (2/3 seeds improve val NLL by 5.7% and 6.9% with confidence; seed 2 by 2.6%, CI crossing zero). The steam-only placement and the latent block are REJECTED against it. Under the amended budget the latent arm converged to the same valley the best seed found before, but not consistently enough to earn its parameters.

Unit	Side-A verdict	Seeds
D-SYN identifiability	SUPPORTED	3/3 (NLL −99 to −136)
O1 posterior, learned	MIXED	−30.1, −31.4, +13.3 (%)
O1 posterior, hybrid	REJECTED	−0.3, −1.6, +3.7 (%)
T1 closure (conservative)	SUPPORTED	+5.7, +6.9, +2.6 (%)
T1 closure (steam-only)	REJECTED	−6.4, −2.3, −3.7 (%)
T1 latent block	REJECTED	−1.6, +0.3, +3.1 (%)
B1 boundary forecaster	REJECTED	CRPS gaps 1.7–2.0
J1 joint training	SUPPORTED	+13.5, +16.4, +33.3 (%)
R1 blindness	SUPPORTED	3/3
R1 leakage (shuffle-controlled)	SUPPORTED	$\Delta = 0.6\%$
R1 direction	v0.2 REJECTED; repair-batch seed 0 tentative PASS	19–34% → 100%

Table 1: Side-A discrimination matrix verdicts. Percentages are validation-NLL improvements (CRPS gaps for B1); confidence intervals per unit as in the committed summary. R1 direction: windows ending negative under a +5% secondary-valve step before (three seeds) and after (seed 0) the repair batch.

B1 boundary forecast. The learned boundary forecaster is REJECTED: all three seeds are significantly worse than persistence on CRPS (gaps of 1.7–2.0, CIs entirely below zero). Boundary channels are strongly autoregressive; a learned forecaster needs more than a GRU to earn its complexity here—a negative result we report as part of the matrix, not around it.

J1 training scheme. Joint training of transition and observer against staged training: SUPPORTED 3/3 (val NLL improvements of 13.5%, 16.4%, 33.3%, all CIs above zero).

R1 action-blind and leakage audit. The closure must not respond to valve channels (runtime blindness check: 3/3 pass) and the boundary whitelist excludes the spray-flow channel. The leakage probe now runs with a shuffle control: an action-aware probe must beat its own shuffled-action twin, not just the action-blind baseline. This rule is itself a product of the matrix. The original probe flagged 23.9% “leakage” on the repaired seed; the shuffle control scored 23.2%, a Δ of 0.6%—the flag was a probe artifact (an underfit 20-epoch budget where any extra input dimension improves validation fit), and the repaired seed passes the gate. The direction probe is reported above: v0.2 failed it (19–34% of windows ending negative), the repair batch corrected seed 0 (100%, -0.228°C per +5%), and seeds 1–2 are pending (<R1-3SEED>).

Each verdict is computed by the frozen runner, not by hand, and every number in this section reproduces from the committed artifacts through the frozen replay script (available on request to qualified reviewers). The matrix doubles as the artifact we propose as an honest benchmark for world-model claims [4]: the proposal is not the checklist but the verdict-repair loop—one gate correction (shuffle-controlled leakage) and one convergence diagnostic (the regime-bias collapse) each changed the protocol or the research agenda, and both are documented in the audit trail. All verdicts are side A only; the side-B matrix is deferred and the single-side scope is carried through the Limitations.

5 Discussion

The repaired physics line holds the action direction that the first training lost, and its terminal-stage accuracy is near persistence; the remaining cost is concentrated and diagnosable. The upstream channel’s one-step error points at the initialization and inversion layer, not the dynamics: anchoring all five observations at initialization (two spray-outlet temperatures currently float) and replacing the fixed pressure-partition assumption with a learned or measured one targets most of the 9.4°C directly (<FIX1-BRANCH>: whether this enters the paper window is decided after the three-seed verdict). Regime-dependent parameter maps, the tempting first remedy, lost their primary evidence once the training budget was corrected—the monotone load bias they were meant to absorb was a training artefact. This ordering illustrates the value of the matrix: one convergence diagnostic and one direction probe re-prioritize the entire optimization agenda.

The magnitude gap deserves a precise status. Direction fidelity is now demonstrated on seed 0; magnitude fidelity is not. The model’s quasi-steady response is $2.7\text{--}7.6\times$ below the nominal mixing

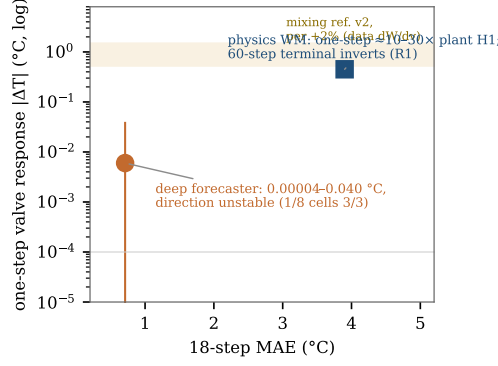


Figure 1: The accuracy–action–fidelity plane (one-step magnitude, log scale). The deep forecaster is accurate but its valve response spans 0.00004–0.040 °C with unstable direction; the physics world model responds on the reference scale at one step (10–30× the plant’s near-zero one-step response); its 60-step terminal response sign-inverted before the repair batch and holds direction on the repaired seed. Shaded band: data-anchored mixing reference for the secondary valve per +2%.

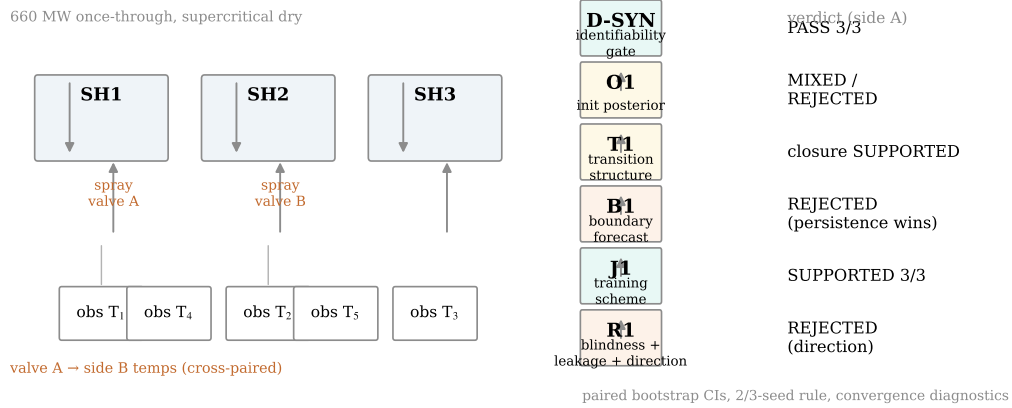


Figure 2: Left: the steam path with five observed temperatures and two cross-paired spray valves. Right: the discrimination matrix with side-A verdicts.

reference, and we argue the reference overstates the plant at the operating point (zero-delay linear mixing against an equal-percentage valve fed by narrow-band excitation). That argument is a hypothesis until the position-binned local-gain probe lands ($\langle CF3 \rangle$); until then we claim direction, not magnitude, for the repaired line. Similarly, the abstract’s promise that the model “knows when it cannot” is exactly what the calibration probe must earn ($\langle D1 \rangle$): the sentence stays only if the interval coverage checks out.

For deployment we endorse the convergent architecture the reduced-order-model lineage articulates [4, 1, 8]: a learned proposer bounded by a certifiable reduced-order model that approves, bounds, or rejects each proposed action within its certified envelope. Planning with imperfect models is a documented failure mode [6, 10]; the two-tier architecture is the constructive answer. The discrimination matrix is the qualification gate for that proposer, and it is, to our knowledge, the first qualification protocol aimed at the setting RL cannot serve: causal action-channel identification from confounded closed-loop records of a safety-class plant, where the only admissible exploration is the plant’s own operation. Forecasting remains necessary; it is not sufficient.

Limitations. Line 1 is teacher-forced, not a recursive foundation forecaster; our claim about recursive forecasters rests on external evidence [9] and should be re-measured with one. Data are from a single unit in dry operation; two-phase regimes are unobserved. The action-response probe covers valve steps at fixed load; larger interventions and load-coordinated moves deserve their own protocol. All

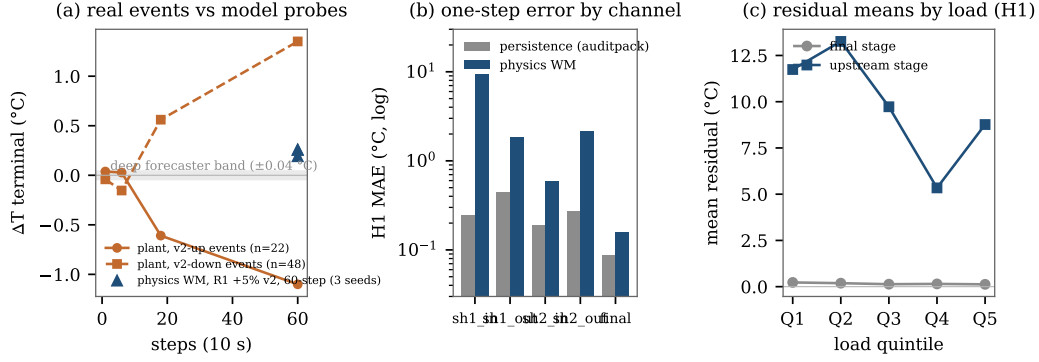


Figure 3: (a) Real secondary-valve events versus model probes: the plant’s response builds over minutes and ends negative on most windows; the physics line’s 60-step terminal response sign-inverted before the repair batch and is sign-consistent after it; the deep forecaster stays inside $\pm 0.04^{\circ}\text{C}$. (b) One-step error by channel: the physics line is near persistence at the terminal stage and $38\times$ worse upstream. (c) Residual means by load quintile: the terminal-stage bias is flat after retraining, the upstream stage swings $5\text{--}13^{\circ}\text{C}$.

matrix verdicts are single-side (A); the side-B matrix is deferred, and no bilateral claim should be read from these results. The quantitative action-response claims use the data-anchored mixing reference, which is identified under closed-loop operation and carries that caveat; magnitude fidelity remains unverified ($\langle\text{CF3}\rangle$), as does interval calibration ($\langle\text{D1}\rangle$). Data and code are not released publicly; the frozen protocol, replay scripts, and committed artifacts are available on request to qualified reviewers.

References

- [1] Peter Benner, Serkan Gugercin, and Karen Willcox. A survey of projection-based model reduction methods for parametric dynamical systems. *SIAM Review*, 57(4):483–531, 2015.
- [2] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. *Advances in Neural Information Processing Systems*, 2018.
- [3] Urban Forssell and Lennart Ljung. Closed-loop identification revisited. *Automatica*, 35(7):1215–1241, 1999.
- [4] Rajat Ghosh. ROM: The mother of world models—verification, physical grounding, and the missing half of the world-model. *arXiv preprint arXiv:2607.03198*, 2026.
- [5] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.
- [6] Taher Jafferjee, Ehsan Imani, Erin Talvitie, Martha White, and Michael Bowling. Hallucinating value: A pitfall of dyna-style planning with imperfect environment models. *arXiv preprint arXiv:2106.05163*, 2021.
- [7] Christopher Rackauckas et al. Universal differential equations for scientific machine learning. *arXiv preprint arXiv:2001.04385*, 2020.
- [8] Lui Sha. Using simplicity to control complexity. *IEEE Software*, 18(4):20–28, 2001.
- [9] Shu Wan et al. Forecast collapse in time-series foundation models. *arXiv preprint arXiv:2608.14106*, 2026.
- [10] Jianhao Wang et al. Double check your state before trusting it: Confidence-aware bidirectional offline model-based imagination. *Advances in Neural Information Processing Systems*, 2022.